Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

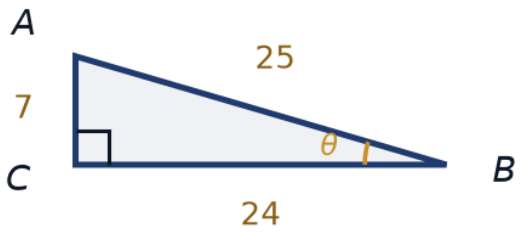
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC studying triangles, $m\angle A = 54^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A culture of bacteria doubles in population every 3 hours. If the initial count is 250 bacteria, write an exponential model $B(t)$ for the population after t hours, and find the population after 12 hours.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find all solutions to $2x^2 - 6x + 5 = 0$ using the quadratic formula, and determine whether the roots are real or complex.

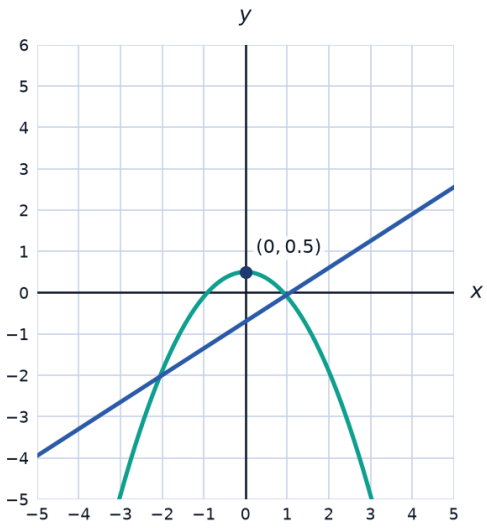
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Quadratic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Functions: Composite & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Functions, Modulus & Quadratic Theory · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

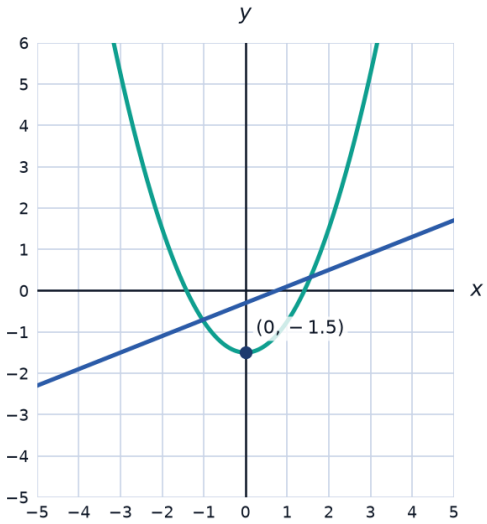
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A solid metal sphere of radius 3 cm is melted down and recast into a cylinder of radius 2 cm. What is the height of the recast cylinder in cm?

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In an arithmetic progression, the 3rd term is 14 and the 7th term is 34. (a) Find the first term a and the common difference d. (b) Find the sum of the first 25 terms.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A 3 kg particle on a rough surface, $\mu = 0.4$. Horizontal force F N. (a) Find the limiting friction. (b) Acceleration when $F = 20$ N. (c) Time to travel 5 m from rest.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Factor the quadratic expression $6x^2 + 11x - 10$ completely.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A circle has radius 8 cm. A sector of this circle has a total perimeter of 28 cm. Find the angle of the sector in radians, and calculate the exact area of the sector.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the set of values of x satisfying the inequality $2x^2 - 7x + 3 \leq 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

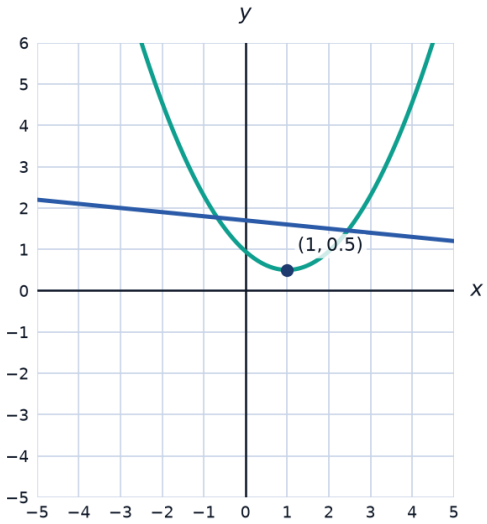
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Geometry

Simplify the algebraic expression $\frac{(2x^3y^{-2})^3}{(4x^{-1}y^4)^2}$ leaving only positive exponents.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Trigonometry

Two parallel lines L_1 and L_2 are cut by a transversal T. A pair of alternate exterior angles are given by $(7x - 14)^\circ$ and $(4x + 31)^\circ$. Find the value of x and the measure of each alternate exterior angle.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Transformations & Vectors

Each interior angle of a regular polygon measures 144°. How many sides does the polygon have?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Probability & Statistics

Two angles $\angle P$ and $\angle Q$ form a linear pair. If $m\angle P = (4x + 12)^\circ$ and $m\angle Q = (2x + 18)^\circ$, find the measure of the smaller angle.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Number

Simplify and rationalize the denominator: $\frac{\sqrt{3}}{\sqrt{5} - \sqrt{2}}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Sets & Venn Diagrams

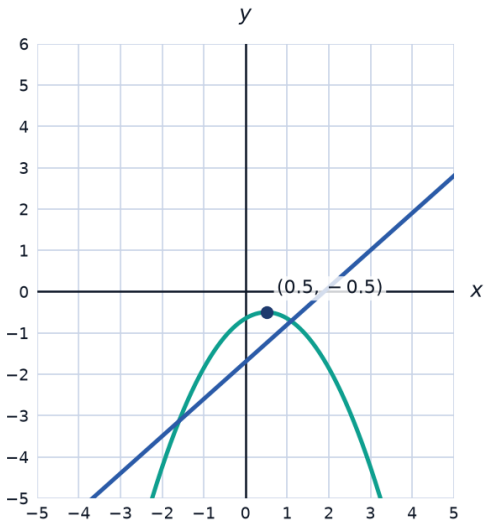
In right ΔABC with right angle at C, altitude CD is drawn to hypotenuse AB. If $AD = 4$ and $DB = 9$, find the length of leg BC.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying geometrical diagrams, $m\angle A = 51^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 84°

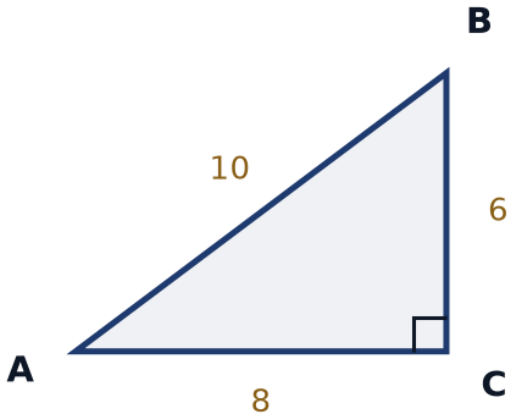
b. 94°

c. 76°

d. 129°

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying measuring angles, $m\angle A = 52^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 83°

b. 93°

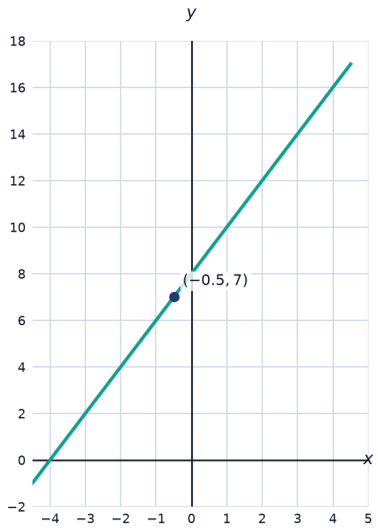
c. 75°

d. 128°

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 8$ in quadrilaterals, evaluate $f(4)$.



a. 16

b. 18

c. 8

d. 20

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the value of x that must be excluded from the domain of $f(x) = \frac{1}{x-6}$.

a. 11

b. 6

c. 1

d. 15

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the smallest integer x for which $f(x) = \sqrt{x-6}$ is defined.

a. 6

b. 0

c. 13

d. -3

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Given that $x = 3.6 \times 10^5$ and $y = 4.8 \times 10^4$, find the value of $x + y$ in standard form.

a. 4.08×10^5

b. 8.4×10^5

c. 4.08×10^9

d. 3.648×10^5

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Write 0.00045 in standard form.

a. 4.5×10^{-4}

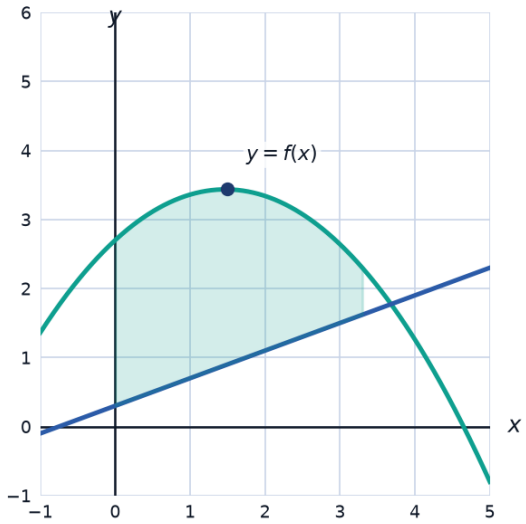
b. 4.5×10^{-5}

c. 45×10^{-5}

d. 4.5×10^{-3}

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The length $L = 12$ cm and width $W = 8$ cm of a rectangle are each measured correct to the nearest centimeter. What is the UPPER bound for the area?



a. 106.25 cm²

b. 96 cm²

c. 117.00 cm²

d. 104.25 cm²

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The ratio of boys to girls in a school is (4 : 5). There are 180 girls. How many students are there in total in the school?

a. 324

b. 144

c. 225

d. 360

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30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A car travels a distance of 140 km, correct to the nearest 10 km, in a time of 2.0 hours, correct to the nearest 0.1 hour. What is the LOWER bound of the average speed?

a. 65.85 km/h

b. 70.00 km/h

c. 74.36 km/h

d. 67.50 km/h

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A jacket is sold for \68 after a 15% discount. What was the original price of the jacket?

a. *x*78.20

b. *x*80.00

c. *x*82.50

d. *x*85.00

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A length is measured as 14 cm correct to the nearest centimetre. What is the lower bound of the length?

a. 13 cm

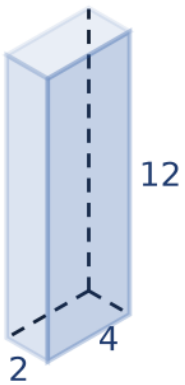
b. 13.4 cm

c. 13.5 cm

d. 13.9 cm

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Two similar geometric solids have a linear scale factor of 1 : 2. If the smaller solid has volume 50 cm³, what is the volume of the larger solid?



a. 400 cm³

b. 200 cm³

c. 100 cm³

d. 500 cm³

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Convert $8000\ \text{cm}^3$ into litres.

a. 8 litres

b. 80 litres

c. 0.8 litres

d. 800 litres

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Convert 35% to a fraction in lowest terms.

a. $\frac{35}{100}$

b. $\frac{7}{20}$

c. $\frac{7}{10}$

d. $\frac{3}{5}$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A circle has an area of $49\pi\ \text{cm}^2$. What is its circumference in terms of π?

a. 14π cm

b. 7π cm

c. 28π cm

d. 21π cm

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A solid hemisphere has volume $18\pi\ \text{cm}^3$. What is its radius?

a. 3 cm

b. 6 cm

c. 9 cm

d. 2 cm

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Probability & Statistics

Divide 420 in the ratio 2 : 3 : 5. What is the value of the largest share?

a. 210

b. 126

c. 84

d. 250

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Number

A solid cylinder has radius 4 cm and height 10 cm. Find its total surface area in terms of π.

a. 112π cm^2

b. 80π cm^2

c. 96π cm^2

d. 160π cm^2

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Sets & Venn Diagrams

Calculate the total value of an investment of 2000 after 3 years at an interest rate of 4% per year compound interest.

a. 2249.73

b. 2240.00

c. 2299.73

d. 2080.00

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Algebra & Graphs

Calculate the volume of a sphere with radius 3 cm in terms of π.

a. 36π cm^3

b. 12π cm^3

c. 108π cm^3

d. 27π cm^3

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Coordinate Geometry

A cone has base diameter 12 cm and vertical height 8 cm. What is its volume in terms of π?

a. 96π cm^3

b. 288π cm^3

c. 48π cm^3

d. 192π cm^3

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Mensuration

A trapezium has parallel sides of length 7 cm and 11 cm, and a perpendicular height of 6 cm. What is its area?

a. 54 cm²

b. 108 cm²

c. 42 cm²

d. 66 cm²

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Geometry

Percentage increase from 40 to 50:

a. 10%

b. 20%

c. 25%

d. 50%

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Trigonometry

In a sale, all prices are reduced by 20%. The sale price of a jacket is 84. What was the ORIGINAL price of the jacket?

a. 105

b. 101

c. 115

d. 100

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